SAECOIN — TECHNICAL GLOSSARY

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Purpose: Define technical terminology used throughout SAECOIN documentation and development.

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1. BLOCKCHAIN FUNDAMENTALS

Blockchain

A distributed digital ledger that stores transactions in cryptographically linked blocks.

Block

A data structure containing a block header and a collection of transactions.

Block Header

The metadata section of a block. It typically contains the previous block hash, Merkle root, timestamp, difficulty target and nonce.

Block Height

The position of a block in the blockchain. The genesis block is generally height zero.

Genesis Block

The first block of a blockchain. It establishes the initial chain identity.

Previous Block Hash

The hash of the preceding block, used to connect blocks together.

Block Hash

A cryptographic digest identifying a block according to its defined hashing rules.

Distributed Ledger

A ledger replicated and maintained by multiple independent network participants.

Full Node

Software that independently downloads and validates blockchain data according to consensus rules.

Network

The collection of connected nodes communicating using the SAECOIN protocol.

Mainnet

The production blockchain network where SAC has its intended real-world economic use.

Testnet

A separate network used for development and testing without relying on mainnet assets.

Fork

A situation where the blockchain history or protocol rules diverge.

Chain Reorganization (Reorg)

A process where a node switches from one valid chain history to another according to the chain selection rules.

Consensus

The rules and mechanisms that allow independent nodes to agree on the valid blockchain history.

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2. PROOF OF WORK AND MINING

Proof of Work (PoW)

A consensus mechanism requiring miners to perform computational work to produce valid blocks.

Miner

A participant or software process that searches for valid block solutions.

Mining

The process of searching for a valid proof of work and proposing new blocks.

Hash Rate

The number of hash calculations performed per second.

Common units:

- H/s — Hashes per second.
- KH/s — Kilohashes per second.
- MH/s — Megahashes per second.
- GH/s — Gigahashes per second.
- TH/s — Terahashes per second.

Nonce

A value that miners vary while searching for a valid block hash.

Difficulty

A measure of how difficult it is to find a block satisfying the network's proof-of-work target.

Difficulty Adjustment

A protocol mechanism that changes mining difficulty according to defined network conditions.

Target

The maximum permitted numerical hash value for a valid proof of work.

Block Reward

The subsidy assigned to a valid block under the monetary rules.

Block Subsidy

New SAC created through the protocol's block reward schedule.

Transaction Fee

An amount paid by a transaction's sender and generally collected by the miner who includes the transaction in a block.

Halving

A scheduled reduction in block subsidy.

ASIC

Application-Specific Integrated Circuit: hardware designed for a particular computational task.

ASIC Resistance

A design objective intended to reduce the performance advantage of specialized mining hardware. It cannot be guaranteed permanently.

CPU Mining

Mining using a computer's central processing unit.

GPU Mining

Mining using a graphics processing unit.

Solo Mining

Mining independently without participating in a mining pool.

Mining Pool

A group of miners combining computational resources and distributing rewards according to pool rules.

Stale Block

A valid block that is not part of the currently selected best chain.

Orphan Block

A block whose parent is not currently known to a node. The precise meaning can vary between implementations.

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3. TRANSACTIONS AND UTXO

Transaction

A signed data structure that transfers value or performs another permitted operation under the protocol.

UTXO

Unspent Transaction Output. An output created by a transaction that has not yet been spent.

UTXO Model

A transaction accounting model in which transactions consume previous outputs and create new outputs.

Transaction Input

A reference to an existing output that a transaction intends to spend.

Transaction Output

A new output created by a transaction, specifying an amount and spending conditions.

Double Spend

An attempt to spend the same output more than once.

Transaction ID (TXID)

A cryptographic identifier calculated from a transaction according to the protocol's serialization and hashing rules.

Transaction Signature

Cryptographic evidence authorizing the spending of an output.

Mempool

A node's temporary collection of valid transactions that have not yet been confirmed in a block.

Confirmation

A transaction is considered confirmed when it is included in a valid block. Additional blocks increase its confirmation count.

Coinbase Transaction

A special transaction that creates the permitted block subsidy and collects applicable transaction fees.

Transaction Fee Rate

The fee paid relative to transaction size or weight.

Dust

A very small transaction output that may be uneconomical to spend under the network's fee policy.

Transaction Serialization

The exact byte representation of a transaction.

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4. CRYPTOGRAPHY

Cryptography

The science of protecting information using mathematical techniques.

Hash Function

A function that converts input data into a fixed-size output.

Cryptographic Hash

A hash function designed to provide properties such as preimage resistance and collision resistance.

SHA-256

A widely used cryptographic hash function that produces a 256-bit digest.

Public Key

A cryptographic key that can be shared publicly and is used in operations such as signature verification.

Private Key

A secret cryptographic key used to authorize transactions or perform other protected operations.

Digital Signature

A cryptographic value proving that a message or transaction was authorized by the holder of a corresponding private key.

Signature Verification

The process of checking whether a digital signature is valid for a message and public key.

Key Pair

A mathematically related public key and private key.

Seed

Secret material used to derive wallet keys.

Mnemonic

A sequence of human-readable words representing recovery material according to a defined standard.

Key Derivation Function (KDF)

A function that derives cryptographic keys from passwords or other secret input.

Encryption

The process of converting readable data into protected data using cryptographic keys.

Decryption

The process of recovering readable data from encrypted data.

Entropy

A measure of unpredictability used when generating cryptographic secrets.

Checksum

A value used to detect accidental errors in data.

Collision Resistance

The property that makes it computationally difficult to find two different inputs with the same hash.

Preimage Resistance

The property that makes it computationally difficult to recover an input from its hash.

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5. WALLET TERMINOLOGY

Cryptocurrency Wallet

Software or hardware that manages keys and creates or signs transactions.

Non-Custodial Wallet

A wallet where the user controls the private keys.

Custodial Wallet

A service where another party controls the private keys on behalf of the user.

Receiving Address

An identifier used to receive SAC.

Change Output

An output returning unused transaction value to an address controlled by the sender.

Wallet Backup

A protected copy of wallet recovery information.

Wallet Restoration

The process of recovering wallet access using backup or recovery material.

Wallet Encryption

Protection of wallet data using cryptographic encryption.

Watch-Only Wallet

A wallet that can monitor addresses and transactions without holding the private keys required to spend funds.

Confirmation Count

The number of blocks counted according to the wallet's confirmation policy for a transaction.

Available Balance

The amount a wallet considers spendable under its current rules.

Pending Balance

The amount associated with transactions that have not yet reached the wallet's required confirmation policy.

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6. PEER-TO-PEER NETWORK

Peer

Another node participating in the network.

Peer-to-Peer (P2P)

A network architecture where participating nodes communicate directly rather than relying entirely on one central server.

Peer Discovery

The process by which a node finds other network participants.

Bootstrap Node

A node used to help new nodes discover network peers.

Seed Node

A network discovery service or node used to provide peer connection information.

Node Synchronization

The process of obtaining and validating blockchain data.

Block Propagation

The process of sharing newly discovered blocks across the network.

Transaction Propagation

The process of sharing transactions between nodes.

Network Partition

A condition where parts of a network cannot communicate with each other.

NAT

Network Address Translation, a technology commonly used by routers to share public IP addresses among devices.

RPC

Remote Procedure Call, a mechanism for software to request operations or information from another process.

API

Application Programming Interface, a defined interface through which software components communicate.

Rate Limiting

Restricting how frequently a client can make requests.

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7. DATABASE AND STORAGE

Database

A system used to store and retrieve structured information.

RocksDB

An embedded key-value database engine. It is a proposed storage option for SAECOIN and has not yet been finalized.

Key-Value Store

A database structure that stores information as key and value pairs.

Chainstate

The current validated state of the blockchain, including relevant unspent transaction information.

UTXO Set

The collection of all currently unspent transaction outputs.

Database Corruption

A condition where stored data becomes inconsistent or unreadable.

Database Recovery

The process of restoring a database to a valid operational state.

Pruned Node

A node that removes certain old blockchain data after validation to reduce storage requirements, according to its pruning design.

Snapshot

A saved representation of blockchain state at a particular point.

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8. SECURITY AND TESTING

Threat Model

A structured description of possible attackers, attack methods, assets and security risks.

Vulnerability

A weakness that could be exploited to compromise software or systems.

Security Audit

A systematic examination of software and its security design.

Fuzz Testing

Testing software with automatically generated or mutated inputs to discover unexpected failures.

Unit Testing

Testing an individual function or small software component.

Integration Testing

Testing whether multiple components work correctly together.

End-to-End Testing

Testing a complete workflow from beginning to end.

Property-Based Testing

Testing general properties across many generated inputs.

Regression Testing

Running previous tests after changes to detect newly introduced problems.

Denial of Service (DoS)

An attack intended to make a system unavailable or degrade its performance.

Sybil Attack

An attack in which one participant creates many identities to influence a network.

51% Attack

An attack involving control of a majority of the relevant mining power in a PoW network, potentially enabling certain chain manipulation attempts.

Reproducible Build

A build process where the same source and build environment produce verifiably identical outputs.

Release Checksum

A cryptographic hash used to verify that a downloaded release file has not changed.

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9. ECONOMIC AND NETWORK METRICS

Maximum Supply

The maximum number of coins permitted by the monetary rules.

Circulating Supply

An estimate of coins available in circulation under a defined accounting methodology.

Total Issuance

The total amount of coins created through permitted issuance rules.

Inflation

An increase in the amount of currency in circulation over time.

Block Time

The target or average time between blocks, as defined by the protocol.

Network Hash Rate

An estimate of the combined mining computation performed by the network.

Mining Difficulty

A measure of how demanding the proof-of-work target is.

Market Capitalization

A market-based valuation commonly calculated using price multiplied by circulating supply. It is not a measure of guaranteed value.

Liquidity

The ability to buy or sell an asset without causing a substantial price change.

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10. SOFTWARE DEVELOPMENT

Rust

A systems programming language intended to provide memory safety and performance.

Cargo

Rust's package manager and build tool.

Crate

A Rust compilation unit or package.

Workspace

A Cargo feature for managing multiple related Rust packages together.

Kotlin

A programming language used for Android application development.

Jetpack Compose

A modern declarative UI toolkit for Android.

Tauri

A framework for building desktop applications using web technologies and a Rust backend.

TypeScript

A typed programming language built on JavaScript.

React

A JavaScript library for building user interfaces.

Git

A distributed version control system.

GitHub

A platform for hosting Git repositories and collaborating on software projects.

CI/CD

Continuous Integration and Continuous Delivery or Deployment. A set of practices for automatically building, testing and delivering software.

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11. SAECOIN PROJECT-SPECIFIC TERMS

SAECOIN

The proposed name of the independent blockchain project.

SAC

The proposed ticker symbol for SAECOIN. Final usage is subject to name and ticker availability review.

Smallest Unit

The smallest indivisible accounting unit of SAC.

Current Proposal: 1 SAC = 100,000,000 smallest units.

SAC Precision

The proposed decimal precision of eight places.

SAECOIN Node

The software responsible for participating in the SAECOIN network and validating blockchain data.

SAECOIN Wallet

The application used to manage SAC keys, addresses and transactions.

SAECOIN Explorer

A service that displays blockchain information such as blocks, transactions and addresses.

SAECOIN Mining Engine

The software responsible for performing proof-of-work calculations.

SAECOIN Mainnet Parameters

The final consensus and network configuration defining the production SAECOIN network.

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12. DOCUMENTATION RULE

Whenever a new technical term is introduced into SAECOIN documentation:

1. Check whether it already exists in this glossary.
2. Use terminology consistently.
3. Add new important terms when necessary.
4. Avoid using one term to describe multiple different technical concepts.
5. Update this document when protocol terminology changes.

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13. IMPORTANT NOTICE

This glossary explains terminology. It does not finalize any protocol parameter or technical implementation decision.

For actual implementation, the relevant approved specifications take precedence.

Where a technical term has multiple meanings across blockchain projects, SAECOIN's own protocol specification must define its exact meaning.